NLP CSC4233

NLP Project Ideas for Undergrads

Bengali · Low-Resource Languages · Product-Focused · Startup-Ready · LLM-Assisted Vibe Coding

Bengali NLP | Startup Ideas | Research Track | Social Impact

20 Projects | Bengali Focus | Vibe Coding | Demo Required | Startup Thinking | Grading Rubric

Project Guidelines — What Is Expected of You

You WILL use LLMs (Claude, ChatGPT, Gemini) and vibe coding tools. That is expected and encouraged. The bar is therefore SET HIGHER — you are expected to understand, explain, modify, and improve what you build.

Understand the Algorithm

You must be able to explain every algorithm you used: why that model, what it does internally, what its loss function is, how it was trained. If you can't explain it to a 10-year-old AND a professor, you don't understand it yet.

Compare Against a Baseline

Your project MUST include a comparison table. Pick 2+ existing methods or baseline models (even TF-IDF + logistic regression counts) and report Accuracy / F1 / BLEU / ROUGE on the same test set. Show your model's improvement (or honest failure).

Modify Something Meaningfully

Don't just fine-tune BERT and call it a day. Add a data augmentation strategy, design a custom loss function, ensemble two models, build a domain-specific pretraining corpus, or add a novel feature. The modification is what earns top marks.

Live Working Demo

Build a web or mobile interface. Streamlit, Gradio, React, or Flutter — your choice. It must accept real input and return real output in real time during your presentation. No screenshots. No 'it worked on my machine'.

Think Like a Founder

For every project: Who is the user? What problem does this solve? How would you monetise it? What would it take to scale to 100,000 users? Include a 1-slide 'Product Vision' in your presentation even if you never build it.

Document Everything

GitHub repo with README, requirements.txt, data pipeline, model card (dataset, architecture, hyperparams, eval results). Reproducible: reviewer must be able to run your demo with one command. Clean code matters.

SECTION 01

Bengali NLP Projects (01–07)

Low-resource challenges, real social impact, and publishable research

Projects 01–03 — Bengali Language Core

01 BanglaSentiment Pro — Multi-domain Sentiment & Emotion Analyser

BENGALI + STARTUP

Problem: Sentiment in Bengali is severely under-studied — existing models fail on social media Banglish, sarcasm, and dialect. Reviews on Daraz, Shajgoj, Chaldal, and Pathao are almost entirely in Bengali yet no production-grade sentiment tool exists.
What to build: Scrape 50K+ reviews from Bangladeshi e-commerce sites. Fine-tune BanglaBERT on 6 emotion classes (joy, anger, fear, sadness, surprise, disgust) + overall sentiment. Add Banglish handling via transliteration. Build a browser extension that overlays sentiment on any Bengali text.

Modifications: Multi-task learning (emotion + sentiment simultaneously). Contrastive loss for sarcasm detection. Custom Banglish normaliser using phonetic mapping.

Stack: BanglaBERT · HuggingFace · Streamlit · FastAPI · Chrome Extension

Product vision: 'Sentiment API for Bangladeshi e-commerce' — charge Daraz/Chaldal $0.001/call for real-time review scoring. 10M calls/month = $10K MRR starting point.

02 BanglaFakeNews Detector — Misinformation Shield for Bangladeshi Social Media

Build a BERT+XGBoost ensemble that classifies Bengali news as real/fake/satire. Extend BanFakeNews dataset by crawling Bangladeshi fact-checking sites (Boom Bangladesh, Rumor Scanner). Add claim extraction and source credibility scoring. Deploy as a Telegram bot for rapid fact-checking.

Stack: BanglaBERT + XGBoost ensemble · Telegram Bot API · MongoDB
Mod: Add claim-evidence attention, multi-hop reasoning across 2 docs
Impact: Social problem — fake news spreads fast in BD WhatsApp groups

03 Bangla Grammar & Spell Checker — the Grammarly for Bengali

No production-grade Bengali spell/grammar checker exists. Build one using a fine-tuned T5 Bengali noisy-channel model. Train on error-injected Bengali Wikipedia text. Deploy as a Word/Google Docs add-in and a REST API. Target: schools, newspapers, government offices.

Stack: Bengali T5 · FastAPI · React · Google Docs Add-on
Mod: Morphology-aware error types · Context-sensitive correction
Product: SaaS for BD government & newspaper chains

Projects 04–07 — Bengali Applications

04 BanglaQA — Open-Domain Question Answering over Bengali Wikipedia

STARTUP

Build a retrieval-augmented QA system: DPR (Dense Passage Retrieval) + BanglaBERT reader. Build your own Bengali Wikipedia passage index with FAISS. Users type a question in Bengali; system retrieves the most relevant passage and extracts the answer span. Demo: a 'Bangla Google' for factual questions.

Modification: domain adaptation for legal (BRTA regulations, labor code) and medical (DGDA drug database). Cross-lingual: accept English questions, answer in Bengali.
BanglaBERT + DPR + FAISS + Streamlit + FastAPI

05 BanglaHate — Hate Speech & Cyberbullying Detector for Bengali Social Media

SOCIAL IMPACT

Scrape and annotate 20K Bengali Facebook/Twitter posts for hate speech targeting religion, gender, and ethnicity. Fine-tune BanglaBERT with focal loss (handles class imbalance). Add span-level explanation (which words triggered the flag). Partner with civil society orgs for annotation quality.

Modification: Multi-label (religion/gender/political hate separately). Cross-script hate spanning Bangla+Banglish. Explainability via LIME/SHAP.
BanglaBERT · SHAP · FastAPI · React dashboard · Meta Graph API

06 BanglaNMT — Low-Resource Neural Machine Translation with Back-Translation

RESEARCH

Train a Bangla↔English NMT model from scratch (or fine-tune MarianMT) using publicly available parallel data. Implement back-translation: generate synthetic Bengali from English Wikipedia to 10x your training data. Evaluate on the WMT Bengali benchmark. Target dialect: Chittagonian or Sylheti as bonus.

Modification: Curriculum learning (easy→hard sentence pairs). Integrate BPE-dropout for robustness. Multilingual pivot (Bangla↔Hindi↔English).
MarianMT + SentencePiece + sacrebleu + Gradio + HuggingFace Hub

07 BanglaSpeech — Speech-to-Text + TTS Pipeline for Bangla

PRODUCT

Fine-tune OpenAI Whisper on 500+ hours of Bengali audio (Common Voice, public radio, BUET corpus). Build a text-to-speech module using a fine-tuned Coqui TTS on Bengali. Deploy as a mobile-friendly progressive web app. Use case: voice search for feature phone users (major BD market segment).

Modification: Dialectal ASR (Sylheti model). Custom noise-robust preprocessing for call-centre audio. Speaker diarisation for meeting transcripts.
Whisper-large-v3 · Coqui TTS · React PWA · WebRTC · Docker

SECTION 02

Product & Startup Projects (08–13)

Build something people will pay for — real users, real problems, real revenue

Projects 08–10 — AI Products with Revenue Potential

08 LegalEase BD — AI Legal Assistant for Bangladeshi Law (Bengali + English)

STARTUP ★★★

Bangladesh has 50M+ cases pending. Most people can't afford lawyers. Build an LLM-powered legal assistant fine-tuned on Bangladesh Penal Code, Family Law, Labor Law, and Land Registration Act. Users describe their situation in plain Bengali; the app explains their rights, what forms to file, and next steps. RAG over legal PDF corpus.

Modification: Multi-document RAG with legal citation tracking. Confidence scoring (model says 'consult a lawyer' when uncertain). Bengali → formal legal language generation.
Stack: LLaMA-3 fine-tuned + LangChain RAG + Chroma DB + React + FastAPI

Revenue: B2C subscription ($2/month), B2B for law firms ($99/month). Addressable market: 50M+ legally underserved Bangladeshis.

09 DocReader BD — Intelligent Document Processing for Bangladeshi Enterprises

STARTUP ★★★

Extract structured data from Bengali invoices, NID cards, land deeds, bank statements, and government forms using OCR + NER. Build a pipeline: image → Tesseract/EasyOCR → LayoutLM (document layout model) → extracted entities in JSON. Target: banks, insurance companies, HR departments for KYC and document verification.

Modification: Fine-tune LayoutLMv3 on Bengali document layouts. Custom NER for BD-specific entities (Thana, Union, Upazila, NID format). Active learning for user corrections.
Stack: LayoutLMv3 · EasyOCR · FastAPI · React Dashboard · PostgreSQL

Revenue: Enterprise SaaS — charge banks $0.10/document processed. BRAC, Dutch-Bangla Bank, Eastern Bank are target customers.

10 AgriBot — Bangla-Language Agricultural Advisory Chatbot for Farmers

STARTUP ★★

60% of Bangladesh's workforce is in agriculture. Farmers have no access to real-time advisory. Build a WhatsApp chatbot (Twilio API) that accepts Bengali text/voice questions about crop disease, weather-based planting advice, fertiliser dosing, and pest identification. Integrate with weather API and DAE (Dept of Agricultural Extension) knowledge base.

Modification: Multimodal — farmer sends photo of diseased leaf → ResNet50 disease classifier → Bengali explanation. Voice input via Whisper.
Stack: Whisper · BanglaBERT · Twilio WhatsApp API · ResNet50 · FastAPI

Revenue: Government (DAE) contracts, NGO partnerships (BRAC, CARE), micro-subscription from farmer cooperatives.

Projects 11–13 — More Product Ideas

11 EduTutor BD — Personalised AI Tutor for SSC/HSC Students in Bengali

PRODUCT

10 million students sit SSC/HSC exams annually. Private tuition is unaffordable for many. Build a Bangla-language adaptive tutoring system: question → student's answer → GPT-4 feedback in Bengali → follow-up question based on weakness. Subjects: Physics, Chemistry, Math. Use Bloom's Taxonomy for question difficulty scaling.

Modification: Curriculum-aware — map questions to NCTB syllabus chapters. Student progress dashboard for parents. Retrieval from NCTB textbook PDF corpus.
Stack: GPT-4 API + LangChain + NCTB RAG + React Native + Supabase

Revenue: $3/student/month subscription. 1% of SSC cohort = 100K students = $300K MRR. Partner with schools.

12 MindBridge — Mental Health Chatbot in Bengali (Safe AI Companion)

SOCIAL IMPACT

Mental health resources are stigmatised and scarce in Bangladesh. Build an empathetic Bengali chatbot trained on therapeutic conversation patterns (CBT, DBT techniques). Integrate PHQ-9 and GAD-7 screening questionnaires. Add escalation: if user is at risk → immediate NIMH hotline connection. Trained on safe, validated psychological frameworks only.

Modification: Emotion detection from text (BERT classifier) gates response strategy. Longitudinal mood tracking. Never diagnose — only support and refer.
Stack: BanglaBERT + GPT-4 system prompt + Twilio SMS + React + crisis API

Revenue: B2B to corporations for employee wellness ($5/employee/month). NGO grants. Telehealth platform partnerships.

13 RecruitBD AI — Bangla Resume Parser and Job Matching Platform

STARTUP ★★★

BD job market: most CVs are in English but candidates describe skills in Bengali. Build a bilingual resume parser using LayoutLM + NER that extracts skills, experience, and education from mixed-language CVs. Add a semantic job-candidate matching engine (cosine similarity over skill embeddings). Target: bdjobs.com alternative with AI ranking.

Modification: Skill ontology built from BD job postings (scrape bdjobs). Salary prediction from experience + skills + location. Bias audit — gender-neutral matching.
Stack: LayoutLMv3 + SpaCy NER + Sentence-BERT + FastAPI + Next.js + PostgreSQL

Revenue: SaaS to HR departments ($99/month unlimited parsing). Premium candidate accounts ($2/month). Job board listings.

SECTION 03

Research Track Projects (14–17)

Publishable results — novel methods, new benchmarks, open-source contributions

Projects 14–17 — Research & Benchmark Building

14 BanglaGLUE — Build a Multi-Task NLU Benchmark for Bengali

RESEARCH / PUBLISH

No GLUE-equivalent benchmark exists for Bengali. Create a benchmark suite with 5 tasks: NLI, sentiment, QA, NER, and grammaticality judgement. Annotate using trained linguistics students + LLM-assisted labelling + expert review. Evaluate 5 models (mBERT, XLM-R, BanglaBERT, GPT-4 zero-shot, your own model). Open-source everything on HuggingFace Datasets.

Novel contribution: The benchmark itself is the paper. Document inter-annotator agreement (Fleiss' κ), test-set leakage analysis, difficulty stratification.
Stack: HuggingFace Datasets + evaluate + Python + GitHub Pages leaderboard

15 Code-Switching NLP — Banglish POS Tagging and Named Entity Recognition

RESEARCH

Banglish (Bengali+English mixed) is used by 30M+ urban Bangladeshis daily. Build a code-switching POS tagger and NER system. Annotate 5,000 social media sentences with language ID + POS tags. Train a multilingual model (mBERT fine-tuned with language-ID auxiliary task). First dedicated Banglish NER corpus.

Novel: Language ID as auxiliary task improves POS accuracy by forcing the model to first decide which language each token is in. Character n-gram features for script-mixed tokens.
Stack: mBERT · CRF · HuggingFace · custom tokeniser for mixed script

16 BengaliOCR — High-Accuracy OCR for Historical and Handwritten Bengali Documents

RESEARCH + PRODUCT

Digitise Bangladesh's documentary heritage. Build a two-stage pipeline: document binarisation + deskew (OpenCV) → CNN-LSTM-CTC OCR model trained on synthetic + real Bangla text. Fine-tune TrOCR on printed and handwritten Bengali. Evaluate on National Archives documents. Open-source a CLI tool for researchers.

Novel: Generate synthetic training data using random Bengali text + diverse font rendering (100+ fonts). Domain adaptation from printed → handwritten using CycleGAN-style augmentation.
Stack: TrOCR · OpenCV · Python · Gradio demo · HuggingFace Hub

17 Sylheti/Chittagonian NLP — NLP for an Endangered Dialect

LOW-RESOURCE RESEARCH

Sylheti has 10M+ speakers but zero NLP resources. Chittagonian (Chattogram dialect) has 15M+ speakers. Build the FIRST POS-tagged and NER-annotated corpus for either dialect. Collaborate with linguistics students to collect 2,000 sentences from YouTube videos, folk songs, and radio broadcasts. Train a transfer learning model from standard Bengali.

Novel: This paper can be published — first systematic NLP study of either dialect. Linguistic documentation contributes to endangered language preservation.
Stack: YouTube transcripts + custom annotation tool (Label Studio) + BanglaBERT fine-tuning

SECTION 04

Advanced & Cross-Cutting Projects (18–20)

Multi-modal, agentic, and next-generation NLP systems

Projects 18–20 — Advanced & Multi-Modal NLP

Advanced Track

18 BanglaVision — Visual Question Answering and Image Captioning in Bengali

MULTIMODAL ★★★

No VQA dataset exists in Bengali. Build one: translate COCO captions to Bengali + collect 5K original Bengali image-question pairs from local photographers. Fine-tune BLIP-2 or LLaVA with Bengali language adapters. Demo: user uploads a photo of a street scene in Dhaka and asks 'এখানে কতটি রিকশা আছে?' (How many rickshaws are here?).

Modification: Culturally-specific visual concepts (rickshaw, gamcha, korta, beel) not in COCO — build a BD-specific object detector. Cross-attention alignment between Bengali tokens and visual patches.

Stack: BLIP-2 / LLaVA + Bengali adapters + COCO-Bengali + Gradio

Product: Accessibility tool for visually impaired Bangladeshis (image→Bengali audio description)

19 BanglaAgent — LLM-Powered Research Assistant That Reads Bengali Web & Documents

AGENTIC AI

Build a ReAct-style agent that can: (1) search Prothom Alo / Daily Star Bengali web, (2) read PDFs, (3) summarise in Bengali, (4) answer follow-up questions with cited sources. Uses tool-augmented LLM (Claude/GPT-4 with custom tools). Demo: 'Explain the latest Bangladesh budget in simple Bengali from this 200-page PDF.'

Mod: Multi-step tool chaining. Bangla summarisation with source attribution (hallucination guard). Output as Bengali podcast script.

Stack: LangChain Agents + GPT-4 + FAISS + BeautifulSoup + Gradio

Product: 'Perplexity AI for Bengali' — research assistant for BD journalists, students, researchers.

20 BanglaLLM — Train the First Bengali-Focused 7B Parameter Language Model

OPEN PROBLEM

The ultimate team project (4–5 students). Collect 50GB+ Bengali text (CC-100, Wikipedia, news, books). Fine-tune LLaMA-3-7B on Bengali using QLoRA (4-bit quantisation, fits on 2×A100). Evaluate on BanglaGLUE. Release on HuggingFace under open licence. This is publishable at ACL/EMNLP.

Mod: Instruction-tuned version (Bengali instruct model). Evaluate on cultural knowledge (BD history, literature, law). Safety filtering for Bengali context.

Stack: QLoRA + LLaMA-3 + BitsAndBytes + 2×A100 GPU (Google Cloud free tier)

This one project could define your career. BanglaBERT author Sudipta Kar is now at Amazon AI. Your name on the first Bengali LLM = permanent NLP history.

All 20 Projects — Quick Reference

01 BanglaSentiment Pro — Bengali — Difficulty	02 BanglaFakeNews — Bengali — Difficulty	03 Bangla Grammar Check — Bengali — Difficulty	04 BanglaQA — Bengali — Difficulty
05 BanglaHate Detector — Bengali — Difficulty	06 BanglaNMT — Bengali — Difficulty	07 BanglaSpeech ASR/TTS — Bengali — Difficulty	08 LegalEase BD — Startup — Difficulty
09 DocReader BD — Startup — Difficulty	10 AgriBot — Startup — Difficulty	11 EduTutor BD — Product — Difficulty	12 MindBridge Mental Health — Impact — Difficulty
13 RecruitBD AI — Startup — Difficulty	14 BanglaGLUE Benchmark — Research — Difficulty	15 Code-Switching NLP — Research — Difficulty	16 BengaliOCR — Research — Difficulty
17 Sylheti/Chittagonian NLP — Research — Difficulty	18 BanglaVision VQA — Multimodal — Difficulty	19 BanglaAgent — Agentic — Difficulty	20 BanglaLLM (7B) — Research — Difficulty

Presentation & Grading Rubric

Your presentation is 20 minutes + 10 minutes Q&A. Judges include the instructor plus two external evaluators (industry or research)[If possible]. The bar is set assuming you had LLM assistance — creativity and depth of understanding are what differentiate grades.

Criterion	Points	Exemplary	Weak
Problem Framing & Motivation Is the problem real and important? Is the target user clear? Is there a compelling 'why now' and 'why you'?	10/100	✅ Clear user persona, quantified problem size, cited existing gap in literature or market.	❌ Vague problem statement, no evidence of user research.
Technical Implementation Correct use of algorithms, model architecture, training pipeline, evaluation setup.	25/100	✅ Custom modification to baseline, ablation study, sound evaluation protocol.	❌ Unmodified tutorial code, no evaluation against baseline.
Algorithm Understanding Can you explain your model internals? Why this architecture? What does each component do?	20/100	✅ Explains loss function, attention heads, hyperparameter choices, failure modes.	❌ Reads from slides. Cannot answer basic questions about the model.
Results & Comparison Performance numbers compared to 2+ baselines on a held-out test set.	20/100	✅ Clear tables, error analysis, honest reporting of failures, statistical significance test.	❌ Numbers without comparison, cherry-picked examples, no test set.
Live Working Demo A functional interface that accepts real input and produces real output during presentation.	15/100	✅ Polished UI, handles edge cases, responsive, shows multiple input types.	❌ Screenshots only. 'Technical difficulties.' Runs from notebook only.
Product / Impact Vision Who would use this? How would you make it sustainable? What is the path to 10K users?	10/100	✅ Revenue model, competitive analysis, 1-year roadmap, named potential partners.	❌ No thought about real-world deployment or sustainability.

Recommended Tech Stack — What to Use and Why

NLP / ML Libraries	
HuggingFace Transformers	Load, fine-tune, evaluate BERT/T5/BART/Whisper in 10 lines of code
BanglaBERT (HuggingFace)	csebuetnlp/banglabert — your go-to for Bengali NLU tasks
spaCy + bnlp-toolkit	Bengali tokenisation, lemmatisation, POS tagging pipeline
LangChain / LlamaIndex	RAG pipelines, agent tools, document Q&A in minutes
FAISS / Chroma	Vector similarity search for RAG knowledge bases

Demo / Interface	
Gradio	Build an ML demo in 5 lines. HuggingFace Spaces hosts it free
Streamlit	More complex dashboards. Charts, file upload, session state
FastAPI	REST API backend. Async, auto-docs, production-ready
React / Next.js	If you want a polished product UI. Use v0.dev with AI help
Flutter	Cross-platform mobile app. AI writes most of the UI code

Data & Infrastructure	
Google Colab / Kaggle	Free GPU for training. T4/A100 GPUs available on Kaggle
HuggingFace Hub	Host your models and datasets. Automatic versioning and sharing
Label Studio	Annotation tool for NER, text classification, span labelling
DVC	Data Version Control — track your datasets like code
Docker	Containerise your demo so it works everywhere ('one command')

Evaluation Tools	
evaluate (HuggingFace)	compute BLEU, ROUGE, F1, accuracy, METEOR in one line
sacrebleu	Standardised BLEU for MT. Essential for fair comparison
RAGAS	Evaluate RAG systems: faithfulness, context relevance, answer relevancy
Weights & Biases	Track experiments, log metrics, compare runs visually
SHAP / LIME	Model explanation — show why the model made each prediction

How to Use LLMs Effectively for Your Project

LLMs are your senior engineering co-pilot — not a replacement for your brain. The best results come from students who understand what they're asking the LLM to build, verify the output, and push beyond the template.

1. Plan first

"Ask Claude: 'I want to build a Bengali sentiment analyser using BanglaBERT fine-…"

LLMs are excellent at outlining architecture and suggesting the right tools. Use this to make decisions before writing code.

2. Build iteratively

"'Write me a HuggingFace Trainer script that fine-tunes BanglaBERT for 3-class se…"

Be specific about every technical requirement. Vague prompts produce generic code. Name the exact library, metric, and trick you want.

3. Debug with context

"Paste the full error + your code + the stack trace. 'I'm getting RuntimeError: E…"

Never ask 'why doesn't this work?' alone. Always include the full error, relevant code, and what you've already tried.

4. Push beyond the default

"'The base BanglaBERT fine-tuned model gives 78% F1. Suggest 5 specific technique…"

After the baseline works, use LLMs to brainstorm improvements. Implement and compare 2–3 suggestions — this is where you earn marks.

5. Document and test

"'Write a README.md for this project with: problem statement, dataset description…"

LLMs write excellent documentation. Use them. A well-documented project signals professional engineering quality to evaluators.

6. Never blindly trust

"Independently verify: (a) does the code actually run? (b) are the numbers real? …"

LLMs hallucinate APIs that don't exist, invent paper references, and produce code that looks plausible but silently gives wrong results. YOU are responsible for correctness.

Register your Group First

Project Deadlines and Presentation schedule

April	28, 29
May	3, 4, 5, 6

** Group wise schedule will be finalized upon getting the responses from all the groups in the system

GOOD LUCK

Go Build Something Real.

The best Bangladeshi NLP startup has not been founded yet. It might be you.

Build for real users

Ship something your grandmother could use. If you can't explain it to her, your demo will fail the Q&A.

Fail early, document honestly

A well-documented failure with clear analysis earns more than a working system you don't understand.

Bengali NLP needs you

230M speakers, near-zero tools. The NLP gap between English and Bengali is a solvable problem and it's yours to solve.

LLMs write code, you think

Your competitive advantage in 2025 is not writing code — it's knowing what to build, for whom, and why it works.